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POLICY AT A GLANCE

Accelerating the Blue Revolution to Enhance Fish Production by 2047 while Ensuring the Protection of Aquatic Ecosystems

Agriculture and Allied Sectors in India

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The '**Policy-at-a-Glance**' provides users with curated information for first-level analysis and inter-state comparisons of policy. The document brings together all topic-relevant national and state-level acts, policies, and schemes in one place. Further, it identifies for a reader what policy features are covered in which document in a simple table.

The first section of this document sets out some '*Context and Background*' on the topic; the subsequent sections explain the '*Policy Features*' that have been compared and include a '*Glossary*' of technical terms used; the last section is the 'Table of Insights' that marks out what '*Policy Feature*' has been covered (and on which page) in each of the topic-relevant documents.

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1. Context

India is the third-largest fish producer in the world and the second-largest aquaculture fish producer. The sector contributes about 7% to global fish production and employs over 2.8 crore people (Department of Fisheries, 2022). With its diverse ecosystems and aquatic species, the Indian fisheries sector has the potential to significantly contribute to the country's nutritional needs, drive economic prosperity, and bolster its reputation as a key player in the global seafood market. Estimates suggest that approximately 60% of India's population consumes fish products (Sajeev, 2020). The average annual per capita consumption is about 6.3 kg across the country, while in coastal areas such as Goa and the Andaman and Nicobar Islands, it is as high as 78 kg (Department of Fisheries, 2022). Consumption of fish and fish products is only expected to rise in the coming years owing to growing awareness of their benefits as well as their relative affordability. However, this burgeoning demand poses a significant risk to the ecosystem due to overfishing, pollution, and climate change. In addition, the Indian fisheries sector faces several challenges that hinder its growth and sustainability. These challenges are multifaceted and include social, economic, environmental, and regulatory issues.

The Pradhan Mantri Matsya Sampada Yojana (PMMSY) is the Government of India's flagship initiative for the fisheries sector. It has three distinct goals: 1) Double farmers' incomes; 2) Increase productivity from three tonnes per ha to five tonnes per ha; and 3) Increase domestic per capita consumption from five kg to 12 kg (Department of Fisheries, n.d.). Inherent to achieving these goals is the need to increase production from the fisheries sector. Thus, enhancing fish production while maintaining the health of the ecosystem is a core long-term strategic goal for the fisheries sector in India.

This document collates information on different policy approaches to build a sustainable fisheries and aquaculture ecosystem in India. As India's fisheries and aquaculture landscape continues to evolve, state governments play a pivotal role in shaping policies that impact the production and growth of the sector as well as the livelihoods of its workforce. Understanding the diverse approaches taken by different states is essential for equitable and sustainable development of the sector.

2. Policy Features

To this end, the national and state governments have taken numerous steps to enhance fish production even as fishing-related regulations ensure ecological conservation.¹ However, it is worth noting that the fisheries sector is largely governed by central policies. This section identifies a common set of policy features that documents referred to in Section 4 (Table of Insights) cover. The features are classified under the following two broad aspects of policy: 1) increasing fish production; and 2) regulating fishing.

1. Increasing Fish Production

- **Upgradation of production, harvesting, and handling infrastructure:** India currently has 8,143 fishing vessels, 74 fishing harbours, and 225 fish handling centres that together comprise the production infrastructure. The total capacity of the fish handling centres is estimated to be 2,085 metric tonnes (MT) (Department of Fisheries, 2022). However, given the objectives of the PMMSY, there is an urgent need to upgrade the existing infrastructure to facilitate the harvesting and handling of additional fish products through national- and state-level initiatives. An example of such a policy is the [Odisha Fisheries Policy](#), that calls for the upgradation of existing fishing harbours and handling facilities as well as the development of new ones. It also recommends setting up an Engineering Wing under the aegis of the state Department of Fisheries to undertake the construction and maintenance of the fishing harbours and handling centres in Odisha. The National Fisheries Policy (2020) also covers similar aspects as do the policies of several other states/union territories (UTs).

¹ The analysis in this document is based on publicly available information from government websites, such as those of the Department of Fisheries, Pradhan Mantri Matsya Sampada Yojana and analogous state government websites, as of 28 August 2023 and is accurate to the best of the authors' knowledge.

- **Upgradation of fish processing and cold chain infrastructure:** Limited cold chain infrastructure and inadequate processing facilities are some of the major challenges faced by the fisheries sector. Presently, the fisheries sector has 605 pre-processing plants in the country with an estimated capacity of 11,026 MT and 623 processing plants that can handle up to 35,915 MT of fisheries products. In addition, there are 50 ice plants with a capacity of 1,486 MT (Department of Fisheries, 2022). These capacities need to be bolstered to ensure continued value addition to the sector. The central government's [Fisheries Infrastructure Development Fund](#) is a financial assistance scheme that encourages private entrepreneurship in the development of fisheries infrastructure. With an estimated fund size of more than INR 7,500 crores, the scheme promotes modernisation of inland and marine fisheries infrastructure and post-harvest loss reduction. This aspect is included in central policy initiatives as well as in the policies of a few states/UTs.

Adoption of modern fish harvesting technologies: Modern fish harvesting technologies such as the use of satellite tracking enhance the efficiency of the fishing exercise, allow for increased harvest, and support the safety of fisherfolk. The [Centrally Sponsored Scheme on Blue Revolution: Integrated Development and Management of Fisheries](#) requires institutions to use satellite imagery on large water bodies and schools of fish, as well as the use of the Global Positioning System (GPS) for the safety of fisherfolk. The states of Gujarat and Odisha also make provisions for the adoption of such technologies in their initiatives.

- **Adoption of sustainable fish production techniques:** Sustainable fish production techniques such as organic fish farming, Biofloc systems and Recirculatory Aquaculture Systems promote responsible harvesting, allowing fish populations to reproduce and replenish, ensuring a consistent source of food and income for communities. Organic fish farming involves raising fish in a manner that prioritises environmental sustainability, animal welfare, and the production of high-quality, chemical-free fish. An example of a policy initiative that promotes sustainable fish production techniques is the [Fisheries Policy of Andhra Pradesh](#) which emphasises the uptake of organic fish farming by earmarking INR 5 crores for research and development (R&D). Meanwhile, several central policies and those of a small number of other states have similar provisions.
- **Species diversification, fish breeding, and genetic improvement practices:** Diversification of fish species through fish breeding and genetic improvement practices is crucial for sustainable fisheries management and aquaculture development in India. These practices aim to enhance fish production, improve the quality of fish stocks, and increase the resilience of the fisheries sector. For example, the [PMMSY](#) gives impetus to species diversification and genetic improvement of fish broods by requiring the establishment of a national network of brood banks. Multiple states/UTs also promote species diversification, fish breeding and genetic improvement through state-level initiatives.
- **Fish feed production:** India's rapidly expanding aquaculture sector relies heavily on the availability of quality fish feeds to support the nutritional requirements of farmed fish. Quality control measures are crucial to ensuring the nutritional quality and safety of fish feeds. Feed manufacturers conduct routine testing to monitor factors like protein content, moisture levels, and the absence of contaminants. The [Goa State Mariculture Policy \(2020\)](#) regulates the quality of fish seed by defining acceptable production norms such as physical properties, raw materials, usage of chemicals and antibiotics, etc. This aspect is also included in central policy initiatives and those of numerous other states/UTs.
- **Extension services:** Fishery extension services play a crucial role in disseminating information, providing training and technical support, and facilitating communication between government agencies, research institutions, and fisherfolk. The services include the establishment of training centres, demonstrations of techniques and technologies, and provisions of diagnostic and quality tests. Gujarat's initiative on [Brackish Water Aquaculture](#) offers a five-day training programme for individuals where shrimp farming operations are demonstrated to trainees. Participants receive INR 250 as a daily stipend during the training and the completion certificate entitles them to receive land for shrimp farming from the government. National-level policies and the policies of many other states/UTs have similar provisions to ensure extension services to fisherfolk.

- **Finance and insurance:** The fisheries sector is often exposed to various uncertainties, including weather-related risks, market fluctuations, and unexpected events. Therefore, financial support and insurance for fisherfolk are critical for economic well-being and risk management. Financial instruments include credit schemes and microloans, while insurance schemes include weather-based insurance, health insurance and life insurance among others. For example, Maharashtra's [Scheme for Grant of Relief to Fishermen](#) provides financial relief to inland and marine fisheries against actual loss in calamitous events. In the case of marine fisheries, the scheme provides coverage for damages to engines, boats, and nets, while for inland fisheries, it covers fish seed stocks and supports extensions of loans and leases. The Kerala Fishermen Welfare Board offers disability and death insurance for active fisherfolk in the state through a [group insurance scheme](#). This aspect of extending financial support and insurance services to fisherfolk is present in national-level policies as well as in those of a number of other states/UTs. Fish farmers across the country are also eligible for Kisan Credit Cards (KCC) which support fish farmers with working capital to cover costs associated with fish seeds, feed, fertilisers, labour, etc. Over 1.3 lakh KCCs had been sanctioned for fisherfolk as of March 2023 (Department of Fisheries, 2023).

2. Regulating Fishing

- **Monitoring, control and surveillance:** The effective governance of fisheries in India – through monitoring, control, and surveillance – regulate fishing activities, ensure compliance with fishing regulations, prevent overfishing, and promote sustainable practices. Examples of monitoring include vessel and dockside monitoring while control is carried out through licensing, seasonal closures etc. Surveillance is usually enforced through patrols and community-based reporting. For example, the [National Fisheries Policy \(2020\)](#) proposes chip-based smart registration cards, mandatory use of logbooks, and integration of information technology and satellites to strengthen governance mechanisms. Multiple states/UTs have similar schemes or policies that allow for monitoring, control, and surveillance of the fisheries sector at the state level.
- **Aquatic health management:** Aquatic health management refers to the systematic and comprehensive approach taken to ensure the well-being and sustainability of aquatic ecosystems, including rivers, lakes, ponds, wetlands, and coastal areas. It includes:
 - **Protection of juvenile fishes:** The survival of juvenile fish, which represent the future of fish stocks, is essential for the long-term viability of the fisheries sector. India's fisheries regulations often include size limits for various fish species. These limits prevent the capture of juvenile fish and provide them the opportunity to grow and reproduce before being harvested. Other ways of protecting juvenile fish are mesh size regulations and closed fishing seasons.
 - **Artificial reef installation:** Artificial reefs are installed to increase fish abundance, diversity, and biomass in specific areas. They serve as fish aggregation devices (FADs), attracting various species, including commercially valuable ones.
 - **Protection of essential habitats:** Mangroves, wetlands, coral reefs, seagrass meadows, estuaries and deltas serve as breeding grounds, nurseries, and feeding areas for fish and other aquatic species. Protecting and conserving these habitats is essential for maintaining fish stocks and ensuring the livelihoods of fishing communities. Policy initiatives protect these areas by designating them as 'no fishing zones' through legal frameworks and engaging communities in these efforts.
 - **Conservation of biodiversity:** The conservation of India's aquatic biodiversity is a complex and multifaceted challenge, requiring a combination of regulatory, educational, and community-based approaches. The introduction and proliferation of exotic (non-native) fish species in India can pose several dangers and ecological risks to local aquatic ecosystems; they can outcompete native species, disrupt local food chains, and negatively impact the balance of aquatic biodiversity and thus, the conservation of indigenous species is essential.

To mitigate the dangers of exotic fishes, India has regulatory frameworks in place to govern the import, sale, and cultivation of exotic species. For example, The Bihar State Reservoir Fisheries Policy (2022) includes a conservation strategy that involves seasonal bans to allow for replenishment of fish, protection of endangered fish species, prevention of the capture and trade of juvenile fish, and enforcement of mesh size regulations. Such aspects are covered in the policies of several other states/UTs as well as in central policies. For example, [The Bihar State Reservoir Fisheries Policy, 2022](#) includes a conservation strategy that involves seasonal bans to allow for replenishment of fishes, protection of endangered fish species, banning capture and trade of juvenile fishes and enforcing mesh size regulations. Such aspects are covered in the policies of 11 states/UTs as well as Central policies.

3. Glossary

Biofloc technology: Biofloc technology refers to the use of aggregates of bacteria, algae, or protozoa, held together in a matrix along with particulate organic matter for the purpose of improving water quality, waste treatment and disease prevention in intensive aquaculture systems (National Fisheries Development Board, n.d.-a).

Recirculatory Aquaculture System (RAS): RAS refers to a type of technology where water is recycled and reused after mechanical and biological filtration and removal of suspended matter and metabolites. This method is used for high- density culture of various species of fish, utilising minimum land area and water (National Fisheries Development Board, n.d.-b).

4. Table of Insights Part 1: Policy Initiatives for Increasing Fish Production

 Geography	 Data	 Policy	 Aspects: Regulating Fishing							
	Total fish production in 2021-22 (in Lakh Tonnes) (Department of Fisheries, 2022)	Reference document	Upgradation of production, harvest and handling infrastructure	Upgradation of processing and cold chain infrastructure	Adoption of modern fish harvesting technology	Adoption of sustainable fish production techniques	Species diversification, fish breeding and genetic improvement practices	Enhancing fish feed production	Extension services for fisheries	Finance and insurance for fisherfolks
Centre (National)	162.5	National Fisheries Policy	✓ (p. 20)	✓ (p. 22)	✓ (p. 7)	✓ (p. 12)	✓ (p. 12)	✓ (p. 14)	✓ (p. 35)	✓ (p. 29)
		Central Sector Scheme on Blue Revolution: Integrated Development and Management of Fisheries	✓ (p. 27)	✓ (p. 27)	✓ (p. 20)	✓ (p. 20)		✓ (p. 57)		
		Pradhan Mantri Matsya Sampada Yojana	✓ (p. 4)	✓ (p. 4)	✓ (p. 5)	✓ (p. 1)	✓ (p. 4)			✓ (p. 5)
		Sagarmala Programme	✓	✓					✓	
		Fisheries and Aquaculture Infrastructure Development Fund (FIDF)	✓ (p. 7)	✓ (p. 7)		✓ (p. 7)	✓ (p. 7)	✓ (p. 7)	✓ (p. 7)	✓ (p. 7)
Andhra Pradesh	48.1	Fisheries Policy of Andhra Pradesh	✓ (p. 13)	✓ (p. 14)	✓ (p. 13)	✓ (p. 13)	✓ (p. 17)	✓ (p. 2)	✓ (p. 14)	✓ (p. 15)
Assam	4.2	Andhra Pradesh Aqua-culture Seed (Quality Control) Act, 2006					✓ (p. 8)	✓ (p. 9)		
		Assam Fish Seed Rules						✓ (p. 2)		
		Major Schemes and its Implementation Guideline Under State Owned Priority Development (SOPD) Scheme	✓ (p. 1)				✓ (p. 2)			
Bihar	7.6	Bihar Fish Seed Certification and Accreditation Act, 2018					✓ (p. 4)			
Goa	1.2	Goa State Mariculture Policy	✓ (p. 3)				✓ (p. 5)	✓ (p. 6)	✓ (p. 3)	✓ (p. 12)
Gujarat	8.7	Accident Assistance Scheme for Fishers								✓
		Marine Fisheries Schemes		✓	✓					✓
		Brackish Water Aquaculture Schemes						✓	✓	✓
		Development of Shrimp Aqua Culture Project	✓ (p. 4)	✓ (p. 4)			✓ (p. 4)	✓ (p. 4)		
Jharkhand	2.6	Jharkhand Fisheries Schemes	✓				✓	✓	✓	✓
Karnataka	10.7	Karnataka Inland Fisheries (Conservation Development and Regulation) Act, 1996					✓ (p. 3)			
Kerala	8.3	Kerala Fisheries Extension and Training Scheme: Kerala Fish Landing Centres	✓						✓	
		Scheme: Fisheries Infrastructure in Kerala	✓	✓						
		Scheme: Inland Fish Production (Kerala)	✓				✓		✓	
		KeralaGroup Insurance Scheme For Fishermen								✓
Madhya Pradesh	2.9	New Fisheries Policy and Rights of Fisheries to Three Tiers Panchayat Programme								✓ (p. 5)

Geography	Data	Policy	Aspects: Regulating Fishing							
	Total fish production in 2021-22 (in Lakh Tonnes) (Department of Fisheries, 2022)	Reference document	Upgradation of production, harvest and handling infrastructure	Upgradation of processing and cold chain infrastructure	Adoption of modern fish harvesting technology	Adoption of sustainable fish production techniques	Species diversification, fish breeding and genetic improvement practices	Enhancing fish feed production	Extension services for fisheries	Finance and insurance for fisherfolks
Maharashtra	5.9	Training Of Fisher Youths							✓	
		Scheme for Grant of Relief to Fishermen								✓ (p. 3)
		Providing Infrastructure Facilities On Fish Landing Centers – Crash Program	✓							
		Scheme: Concession In Electricity Bill For Ice Factories And Cold Storages Of Fisheries Co-Operatives		✓						
Manipur	0.3	Scheme: Fishery Research and Extension & Transfer of Technology							✓	
		Scheme: Establishment of Fish Farms						✓		
Meghalaya	0.2	Meghalaya State Aquaculture Mission 2.0					✓ (p. 50)	✓ (p. 31)	✓ (p. 43)	
Odisha	9.9	Odisha Fisheries Policy	✓ (p. 15)	✓ (p. 15)	✓ (p. 14)	✓ (p. 10, 11, 12)	✓ (p. 12)	✓ (p. 13, 28, 29)	✓ (p. 11)	✓ (p. 20)
		Scheme: Input Assistance to develop Network of Fish Seed Growers					✓ (p. 6)		✓ (p. 10)	✓ (p. 8)
		Scheme: Input Assistance to Farmers for Taking up Fish Farming in Farm Ponds								✓ (p. 1)
		Support to Private Fish Hatcheries for Adoption of Early Breeding and YearRound Seed Production Technology and for Hatchery Accreditation and Seed Certification	✓ (p. 1)				✓ (p. 1)		✓ (p. 2)	✓ (p. 1)
		Scheme: Input Assistance to Women Self Help Groups for Pisciculture in Gram Panchayat Tanks	✓ (p. 1)						✓ (p. 3)	✓ (p. 1)
		Scheme: Promotion of Intensive Aquaculture through Introduction of Bio-floc Technology	✓ (p. 1)			✓ (p. 1)			✓ (p. 5)	✓ (p. 1)
		Machha Chasa Pain Nua Pokhari Khola Yojana (Matsya Pokhari Yojana)	✓ (p. 4)						✓ (p. 8)	✓ (p. 5)
		Scheme: Popularisation of Fishery Machinery/ Equipments/ Implements for Intensive Aquaculture in Odisha	✓ (p. 1)							✓ (p. 1)
		Scheme: Development of Brackish Water Fisheries	✓ (p. 1)							✓ (p. 1)
Tamil Nadu	8.1	Fishing Ban Period Assistance Scheme								✓
		Fisheries Management for Sustainable Livelihood- II Project				✓	✓		✓	
		Scheme: Subsidy Assistance of 50% to fishermen towards procurement of Tuna long liners	✓ (p. 3)							✓ (p. 3)
Telangana	3.9	Development of Fisheries Schemes	✓				✓			✓
		Implementation Guidelines for Integrated Fisheries Development Scheme	✓ (p. 1)	✓ (p. 1)			✓ (p. 1)		✓ (p. 1)	
West Bengal	18.4	West Bengal Inland Fisheries Policy	✓ (p. 3)	✓ (p. 8)		✓ (p. 6)	✓ (p. 6)	✓ (p. 6)	✓ (p. 9)	✓ (p. 3)

5. Table of Insights Part 2: Policy Initiatives to Regulate Fishing

 Geography	 Data	 Policy	 Aspects: Regulating Fishing	
			Monitoring, Control & Surveillance for regulating fishing	Aquatic health management
Centre (National)	162.5	National Fisheries Policy Pradhan Mantri Matsya Sampada Yojana Fisheries Policy of Andhra Pradesh	✓ (p. 8)	✓ (p. 27) ✓ (p. 4) ✓ (p. 13)
Andhra Pradesh	48.1	Andhra Pradesh Marine Fishing (Regulation) Act, 1995	✓ (p. 4)	
Assam	4.2	Assam Private Fisheries Protection Act, 1935	✓ (p. 1)	
Bihar	7.6	Bihar State Reservoir Fisheries Policy		✓ (p. 5)
Chhattisgarh	5.9	Fisheries Act 2006	✓ (p. 8)	✓ (p. 8)
Goa	1.2	Goa Marine Fishing Regulation (Amendment) Act, 2019 Goa State Mariculture Policy	✓ (p. 3) ✓ (p. 5)	 ✓ (p. 3)
Gujarat	8.7	Gujarat Fisheries Act, 2003	✓ (p. 4)	✓ (p. 3)
Karnataka	10.7	Karnataka Inland Fisheries (Conservation Development and Regulation) Act, 1996	✓ (p. 4)	✓ (p. 4)
Kerala	8.3	Scheme: Kerala Conservation and Management of Fish Resources	✓	
Madhya Pradesh	2.9	New Fisheries Policy and Rights of Fisheries to Three Tiers Panchayat Programme Madhya Pradesh Fisheries Act, 1948	✓ (p. 7) ✓ (p. 3)	
Maharashtra	5.9	Maharashtra Marine Fishing Regulation Act, 1981	✓ (p. 6)	
Manipur	0.3	Manipur Fishery Rule		✓ (p. 2)
Mizoram	0.1	Mizoram Fisheries (Amendment), Act, 2022		✓ (p. 5)
Odisha	9.9	Odisha Fisheries Policy	✓ (p. 17)	✓ (p. 23)
Tamil Nadu	8.1	Tamil Nadu Aquaculture (Regulation) Act, 1995	✓ (p. 3)	✓ (p. 4)
Telangana	3.9	Development of Fisheries Schemes		✓
		Order: Restriction & Regulation of Exotic Fish Culture & Tilapia (Oreochromis mossabicus) to Conserve the Indigenous Aquatic Diversity in the Water Bodies of Telangana		✓ (p. 1)
Uttar Pradesh	8.1	United Provinces Fisheries Act, 1948	✓ (p. 6)	
West Bengal	18.4	West Bengal Inland Fisheries Policy West Bengal Marine Fishing Regulation Act, 1993	 ✓ (p. 5)	✓ (p. 3)

Note 1: Documents linked in the tables include Acts, Policies, Schemes, Programmes, and Guidelines which facilitate an overview and comparison of central- and state-level initiatives. While all relevant documents have been identified, the coverage may not be exhaustive. Further, the 'Policy-at-a-Glance' is a mapping of these identified documents to highlight aspects of fisheries and aquaculture management.

Note 2: Page numbers specified in the Table of Insights are as per the PDF pagination opened in a Google Chrome browser. It is possible to encounter differences in the actual page number when documents are opened in a different web browser/application.

6. References

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